Rocket Flight Simulator and Mission Analysis Tool

Mission Report

Mission Summary

This report summarizes a one-dimensional vertical launch using variable thrust, propellant mass depletion, altitude-dependent atmospheric density, and velocity-opposed aerodynamic drag.

Rocket Parameters

Parameter	Value
Dry Mass	117000.000
Fuel Mass	663000.000
Thrust	13300000.000
Burn Time	180.000
Drag Coefficient	0.300
Cross Sectional Area	22.900

Atmospheric Model

Density follows a simplified International Standard Atmosphere: 1.225 kg/m^3 at sea level, cooling through the troposphere, then an isothermal lower-stratosphere approximation above 11 km.

Flight Phase Timeline

Phase	Start (s)	End (s)	Duration (s)
PRE-LAUNCH	0.000	17.100	17.100
LIFTOFF	17.200	17.200	0.000
POWERED_ASCENT	17.300	179.800	162.500
COASTING	179.900	300.000	120.100

Performance Metrics

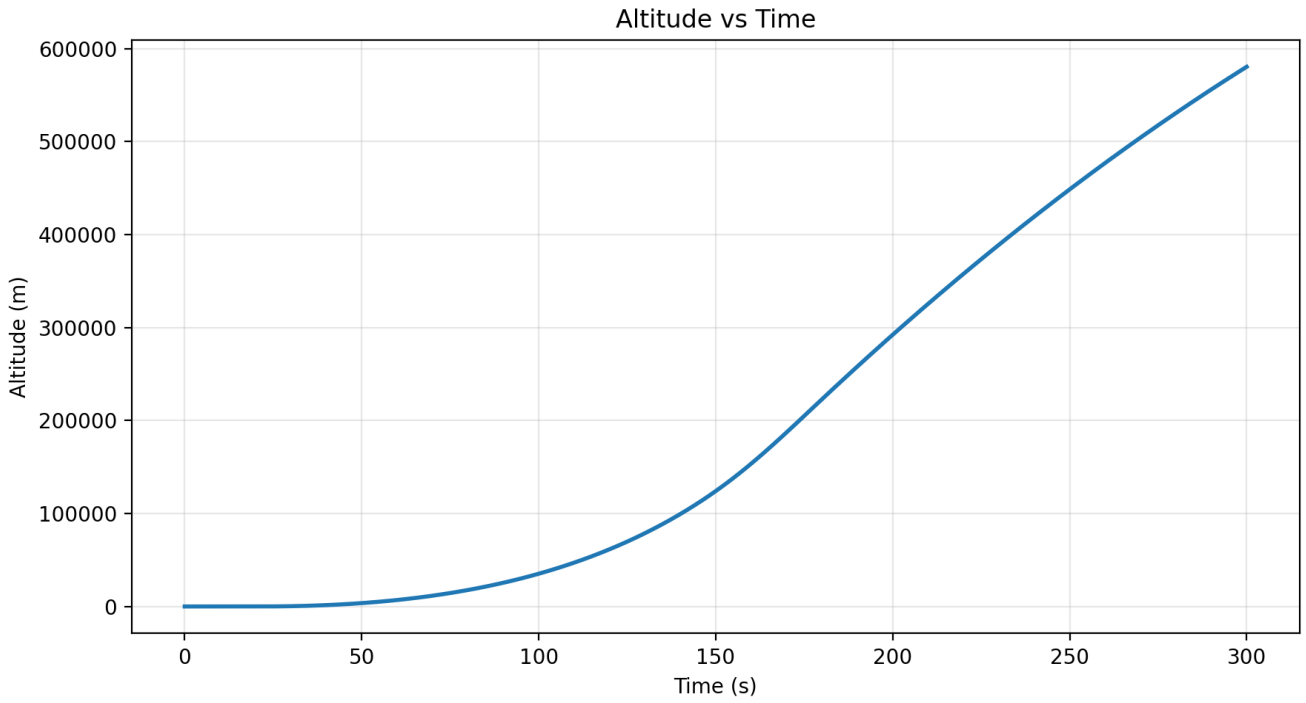
Metric	Value
Maximum Altitude M	580348.693
Maximum Velocity Mps	3584.049
Maximum Acceleration Mps2	52.068
Burnout Time S	179.800

Apogee Time S	300.000
Total Flight Duration S	300.000
Landing Velocity Mps	2391.632
Flight Duration S	300.000

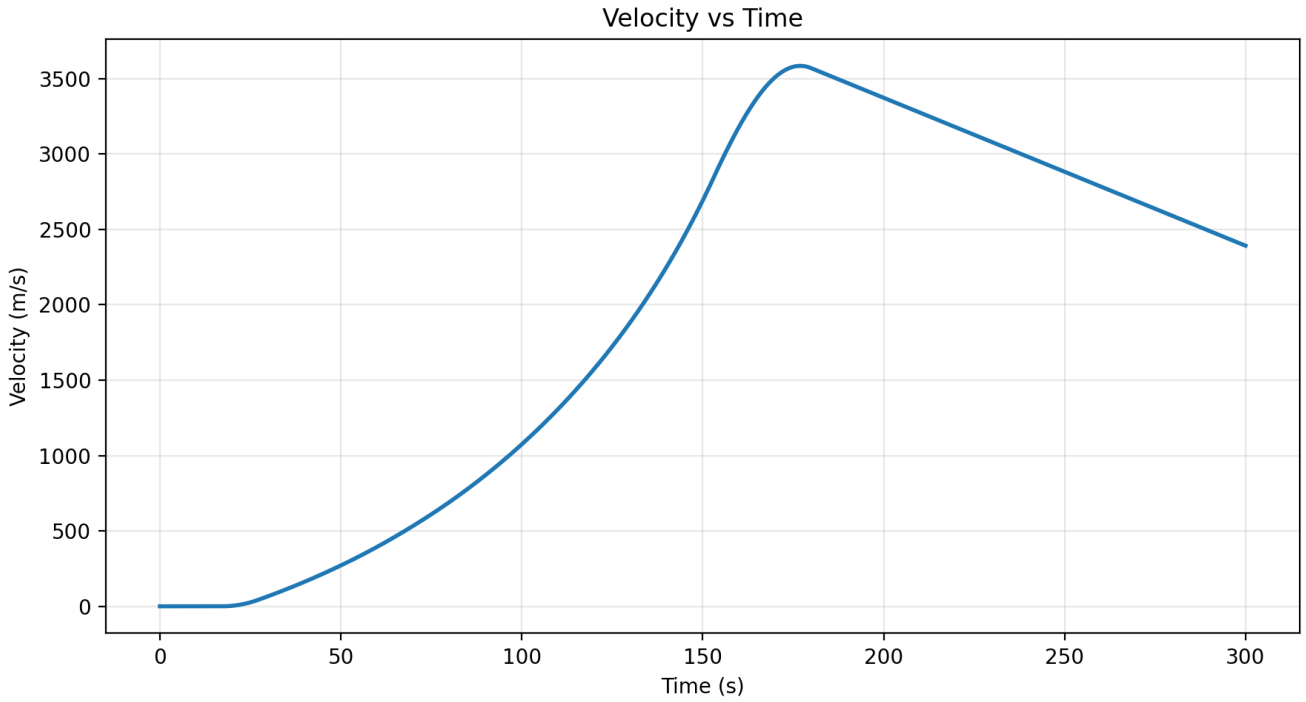
Engineering Conclusions

The upgraded simulation captures the dominant first-order effects for vertical flight analysis: decreasing density reduces drag at altitude, thrust tail-off shapes burnout acceleration, and mass depletion improves the predicted acceleration history.

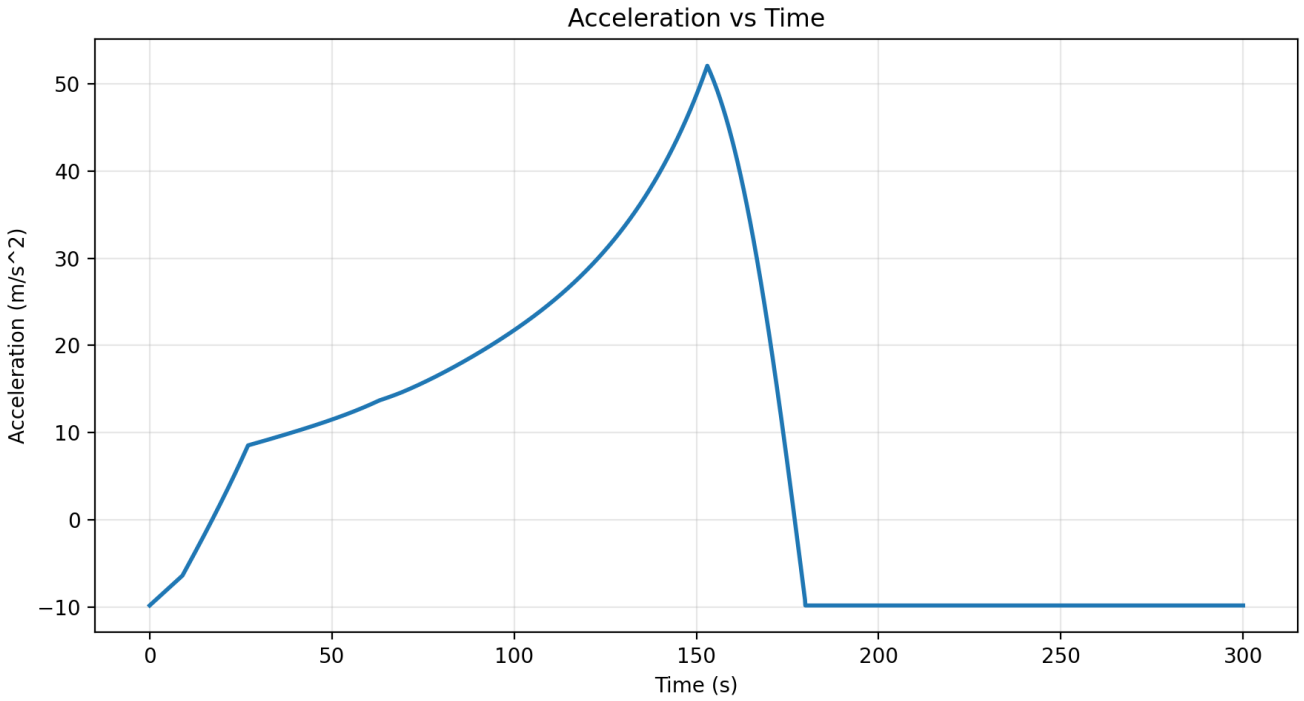
Altitude Plot



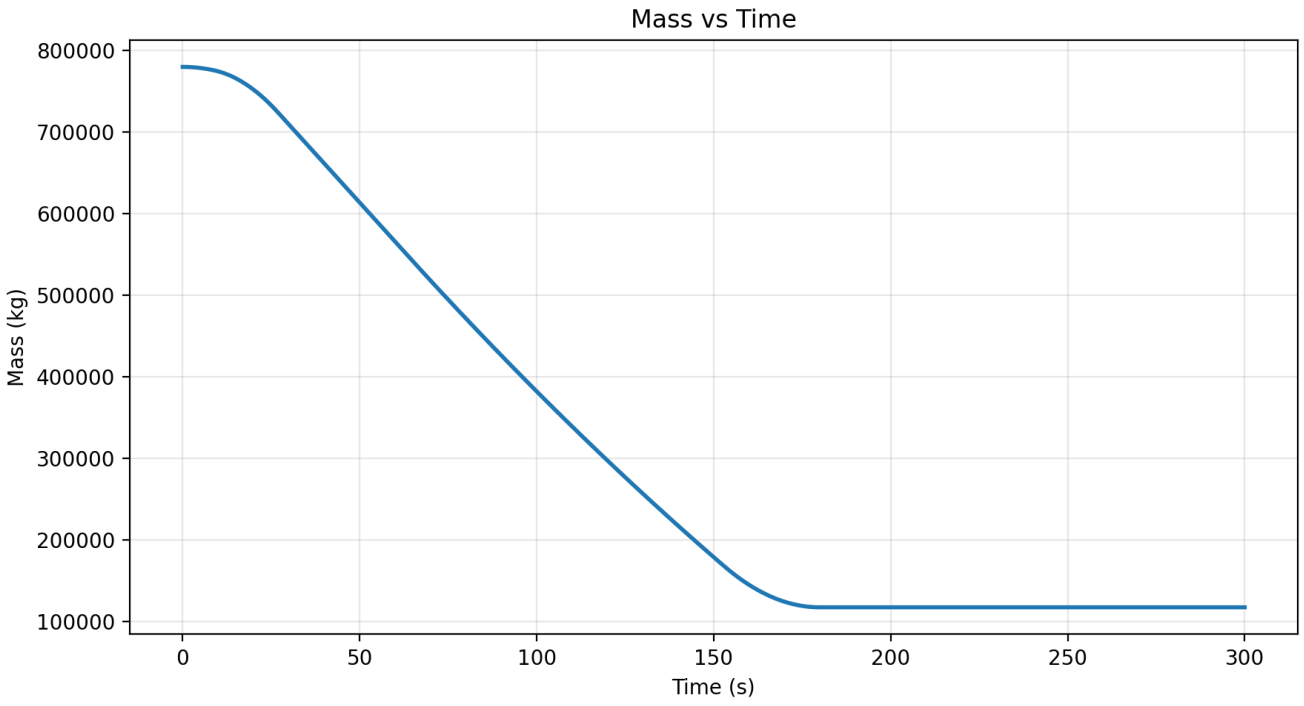
Velocity Plot



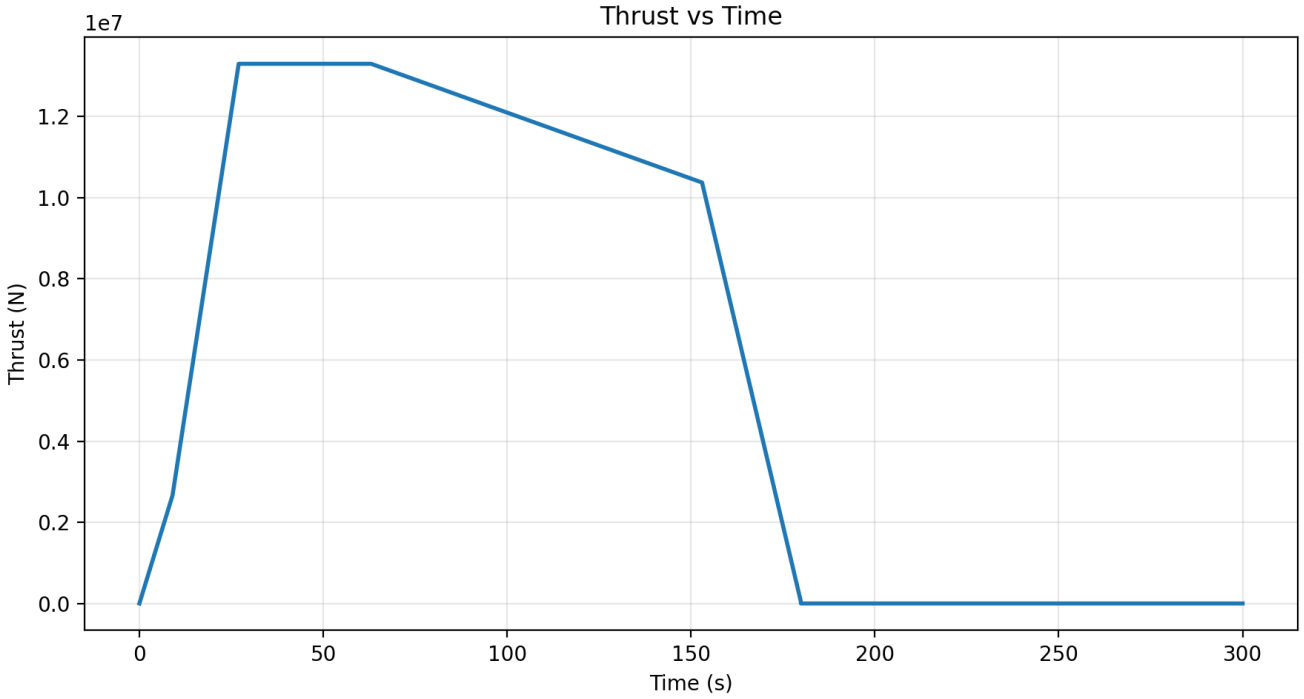
Acceleration Plot



Mass Plot



Thrust Plot


Thrust vs Time
1e7
1.2
1.0
0.8
0.6
0.4
0.2
0.0
Thrust (N)
0
50
100
150
200
250
300
Time (s)